CloudSage — Roadmap and Future Expansion

How the project grows after the first working implementation

Phase 1 — EC2

Focus entirely on EC2 rightsizing and the complete lifecycle from discovery to verification.

Phase 2 — Resource Waste

- Detached EBS
- Unused Elastic IPs
- Old snapshots
- Unused load balancers

Phase 3 — Storage and Database

- S3 lifecycle/storage class analysis
- RDS rightsizing
- RDS idle/underutilized detection

Phase 4 — Compute Platforms

- ECS
- Lambda
- Auto Scaling

Phase 5 — Network Costs

- NAT Gateway
- Data transfer
- Cross-AZ patterns
- Load balancer usage

Phase 6 — Multi-Account

Use AWS Organizations/assume-role patterns to analyze multiple accounts while preserving account isolation and audit trails.

Phase 7 — Platform Engineering

- Web dashboard
- RBAC
- Scheduled scans
- Worker queue
- Notifications
- K3s deployment
- Prometheus/Grafana monitoring

Phase 8 — Intelligence

- Historical outcomes
- Recommendation quality scoring
- Internal runbook RAG
- Natural-language query interface
- What-if cost simulation
- Budget-aware recommendations

Phase 9 — Controlled Autonomy

Only after extensive verification should low-risk actions be eligible for automatic execution. Production and destructive operations remain approval-gated by default.

Future What-If Engine

A user could ask: 'What happens to monthly cost and operational risk if we reduce these EC2 instances by one size?' The system can simulate the configuration change, estimate cost, show Terraform impact and explain assumptions without applying anything.

Long-Term Product Vision

CloudSage becomes an infrastructure intelligence and optimization layer: it understands what exists, what it costs, how it is being used, how it is managed through IaC, what can be safely changed, and whether the change delivered the expected result.